



Model-driven safety and security co-analysis: A systematic literature review[☆]

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ABSTRACT

Failures in systems that can lead to loss of life, property, and environmental damage, make them safety-critical systems requiring the analysis and demonstration of dependability properties. When those systems can connect/disconnect to each other, it raises security issues, making them also security-critical. Safety and security are crucial in safety- and security-critical design. The complexity of a connected world impacts safety and security, requiring dependability assurance processes in compliance with standards like ISO 26262, ISO 21434, IEC 61511, and IEC 61508. Model-Driven Engineering (MDE) plays a significant role by using models to represent and analyze safety and security properties, facilitating their integration. This study aims to explore the role of MDE in supporting safety and security co-engineering through a Systematic Literature Review. We identified 119 studies that were analyzed and categorized based on the life-cycle phase, risk assessment activity, application domain, methodology type (integrated or unified), methodology goal (co-engineering or cross-fertilization), measurement (qualitative, quantitative, or both), modeling approach, representation, evaluation method, and supporting tools. Though model-driven contributions are less prevalent than model-based ones, tools widely support them. Studies primarily focus on transportation (especially automotive) and industrial domains, but there is potential for broader participation with increasing IoT adoption.

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1. Introduction

Safety and security are important quality attributes of any software system (Kenner et al., 2021), especially in the context of safety-critical domains (e.g., automotive, aerospace, manufacturing, and medical devices Knight, 2002), in which faults and failures that cause harm to the system, its users, or to the environment must be avoided. Safety is the absence of unreasonable risk of physical harm to people (serious injuries or loss of life), significant damage to the property, and environmental damage due to system malfunctioning behavior, and its correct operation in response to user's inputs (IEC, 2010).

Security, on the other hand, is the prevention of unauthorized access from malicious external agents aiming to change or damage equipment, information, or services (IEC, 2009). For example, autonomous cars must ensure safe operation, preventing accidents and physical injuries to people (Dajsuren and van den Brand, 2019); cloud computing systems and services connected to the Internet must ensure data confidentiality, integrity, and availability (Dillon et al., 2010); and cyber-physical systems should guarantee the safety of involved humans (Biffel et al., 2019).

Cyber-physical systems (CPS) are smart systems that include engineered interacting networks of physical and computational components (NIST, 2016). CPSs combine physical processes and computer systems, comprising sensors to observe the environment and actuators that modify physical processes (Kopetz et al., 2016). CPSs are often safety-critical since any system malfunction might seriously harm users or the environment. For instance, a malfunctioning pacemaker may harm its patient. CPSs can also be considered security-critical if they communicate with the external world since the communication channel is a vulnerability that attack vectors could exploit against the CPS (Biro et al., 2017).

CPSs in safety-critical domains like autonomous cars have the ability to connect and disconnect to each other at runtime (i.e., openness), and are capable of modifying their behaviors in response to changes in the environment (i.e., adaptiveness) (Wei et al., 2018).

Since CPSs and Internet of Things (IoT) systems are heavily connected (e.g., to the Internet), it raises security issues, making them susceptible to external cyber-attacks. Attack vectors could exploit vulnerabilities to compromise system and/or data confidentiality, integrity,

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availability, or other security properties such as authenticity, authorization, and accountability.

A security vulnerability is an internal system failure caused by faults in the design (e.g., unintended feature interactions), development (e.g., buffer overflow), or operation (e.g., configuration error) (McGraw, 2006). Thus, if the assurance of security properties is imperative, then safety-critical systems can also be security-critical¹. However, safety and security are frequently addressed separately (Kruck et al., 2021; Kenner et al., 2021). While the malfunctioning behavior is addressed by safety engineering, the increasing risk of intentionally caused harm is addressed by security engineering (Lisova et al., 2019). But, with the increased connectivity between heterogeneous systems, safety and security must be integrated to properly address the unreasonable risk of harm due to malfunctioning behavior or malicious intent from external agents (Lisova et al., 2019).

Since separating safety and security risk analysis is no longer feasible (Biro et al., 2017), we need methods that deal with them simultaneously because, in some situations, safety and security requirements may conflict with each other. For example, encrypting a message for security reasons may increase the delivery time to a value greater than the required threshold to address safety. Moreover, existing standards address safety and security separately. For instance, in the automotive domain, ISO 26262 (ISO, 2011) addresses functional safety, while ISO 21434 (ISO, 2021) focus on cyber-security.

Another important concern is that treating safety and security separately and dealing with their integration later at the development stage implies a greater effort to identify inconsistencies between conflicting safety and security requirements and common measures. Thus, the late detection of inconsistencies demands repetitive iterations between safety and security engineering to solve them. The early integration of safety and security aiming to detect issues and trade-offs may reduce the number of iterations to identify common measures (Urretavizcaya et al., 2021). Both disciplines use an acyclic graph to describe the propagation of failures/attacks across a system model. While safety describes fault and error propagation across a system model using Fault Tree Analysis (FTA), security considers attack paths and their links to threats and security properties, like the CIA Triad, which stands for Confidentiality, Integrity, and Availability, and goals using Attack Trees.

The design of critical systems demands the management of interactions between safety and security requirements that can be conflicting. In this context, model-based and model-driven approaches can be employed to support the integrated analysis of security concerns (e.g., expressed in attack trees) and their impact on system safety properties using FTAs. While Model-Based Engineering (MBE) assists the development, Model-Driven Engineering (MDE) is a Software Engineering paradigm in which models are primary artifacts, *driving* the system development (Selic, 2003). The benefits of automation and consistency from MDE have been explored in the last decade in both academia and industry to support the integration and analysis of the interplay between safety and security engineering models.

In this paper, we present a Systematic Literature Review (SLR) (Kitchenham, 2004) to investigate the current state of the art of modeling approaches, namely model-based or model-driven, for safety and security co-analysis. We aim to get a better understanding of existing modeling approaches to support the analysis of the interplay between safety and security. To achieve this goal, we identified the domains where such techniques are applied, their principles, how they were evaluated, and whether they have tooling support.

Based on the analysis of 119 selected publications, our findings revealed several gaps concerning the analysis of safety and security interplay while using modeling approaches. Those gaps include few

model-driven methodologies dedicated to formal modeling; the lack of studies addressing more specific application domains, with a predominant focus on transportation, particularly in automotive and industrial domains; the significant number of approaches that choose safety as a starting point for co-analysis, while the inverse has not received enough attention; and the shortage of tools that natively support safety and security co-analysis. These findings open opportunities for potential directions in future research.

The remainder of this paper is organized as follows. Section 2 presents a brief background on model-driven and model-based approaches, safety, and security concepts needed to better understand the contributions and analysis developed in this study. Section 3 details the planning and execution procedure of our SLR. The SLR results are shown in Section 4. Section 5 discusses the main findings of the SRL results. Section 6 presents the threats to the validity of our study. Section 7 presents related work and compares them to this study, highlighting the differences in methodology and objectives. Finally, in Section 8, we present the conclusions and future research directions.

2. Background

In this section, we provide the key background concepts on model-based and model-driven engineering, safety and security engineering disciplines, formal methods, and the interplay between safety and security necessary for a better understanding of the SLR results.

2.1. Modeling approaches

Models are abstractions from reality that can represent it in a partial or simplified way. In the software industry, *“these models are used to exchange information between engineers to, for example, further refine requirements, implement software/systems/architecture, including software deployment”* (Nigam et al., 2018). In this study, we adopted the modeling approaches classification scheme defined by Brambilla et al. (2012). We distinguish between model-driven and model-based approaches.

Model-driven is an approach where models are used as the primary artifact of the development process. Usually, the resulting implementation is semi or automatically generated from those models. This approach has several acronyms, such as Model-Driven Development, Model-Driven Engineering, Model-Driven Software Engineering, and Model-Driven Architecture. Still, they all share the same line of thought.

Model-Based approaches, on the other hand, are a softer version of model-driven approaches. *“The MBE process is a process in which software models play an important role although they are not necessarily the key artifacts of the development (i.e., they do NOT “drive” the process as in MDE)”* (Brambilla et al., 2012). For the remainder of this paper, aiming to achieve a more consistent and straightforward reading, we use the acronyms: MD for model-driven and MB for model-based approaches.

2.2. Safety

According to Draeger (2015), safety is typically concerned with avoiding adverse effects of the system on its environment. Safety issues are usually related to damage to people, property, and the environment. As mentioned in Section 1, FTA and FMEA are the first techniques for safety analysis, and they date from 1960 (Hecht et al., 2020; Nai Fovino et al., 2009). FTA is a tree-based top-down analysis and the last is a bottom-up supported by spreadsheets. There are also other relevant techniques like System Theoretic Process Analysis (STPA) (Castiglione and Lupu, 2020), which defines the system control structure and identifies its hazardous control actions; Hazard and Operability Analysis (HAZOP) (Friedberg et al., 2017), which uses guidewords over system parameters to identify deviations from the designed behavior; and Hazard Analysis and Risk Assessment (HARA), widely used in the automotive domain, which classifies hazards considering severity, exposure, and controllability (Martin et al., 2020).

¹ From now on, we adopted the *critical systems* term to refer to safety and security-critical systems

2.3. Security

Security issues are mainly related with risk to the availability, confidentiality, and integrity of systems threatened by intentional or unintentional attacks. Those issues can be physical or virtual. As mentioned by [Abdo et al. \(2018\)](#), the increasing usage of the Internet for connecting, remote controlling, and supervising systems and facilities also increases the risk related to cybersecurity.

Security also has its own techniques (e.g. STRIDE [Hernan et al., 2006](#)), some of which were based on safety analog techniques. AT, already mentioned in Section 1, is analogous to FTA. The E-Safety Vehicle Intrusion Protected Applications (EVITA) project proposed the Threat and Operability Analysis based on the well-known HAZOP method, which uses guide-words for the analysis ([Dürrewang et al., 2017](#)). The Threat Analysis and Risk Assessment (TARA) method is based on Common Criteria ([Cybersecurity & Infrastructure Security Agency, 2006](#)) and EVITA, and it is very similar to HARA. The TARA method considers the expertise, time, knowledge, access, and resources needed to make an attack feasible ([Förster et al., 2019](#)).

2.4. Formal methods

Modeling approaches use structural, semiformal, or formal machine-readable models instead of non-structured word processing documents ([Papakonstantinou et al., 2019](#)). According to [Brunel et al. \(2014\)](#), formal techniques (i.e., techniques using formal models), can be used to support safety and security assessment due to their mathematical foundation, which allows them to prove some requirements with higher confidence than testing and manual review. Thus, using formalisms can reduce or even eliminate possible defects in the software, maintaining the system's safety ([Howard et al., 2017](#)). Although we recognize the important role formal methods play in modeling approaches, they are out of the scope of this SLR, and another work should be conducted to address it.

2.5. Safety and security interplay

As discussed in Section 1 and previous Sections 2.2 and 2.3, both disciplines are indispensable for acquiring a completely safe and secure system, although a new challenge emerges when dealing with them in the same software development process. The expertise required for engineers to deal with those disciplines remains split, but the components and connections that are assessed by safety and security analysis are the same. Therefore, an effort must be made to align them.

According to [Sabaliauskaite and Mathur \(2015\)](#), a weak alignment between those disciplines may produce inefficient development and partially-protected systems. We can accomplish excessive costs by defining redundant countermeasures or degradation of the whole safety of the system by adopting mechanisms that increase safety and decreases security at the same time, and vice-versa. Safety and security can have different kinds of interplay:

- **No Interference:** Properties from one discipline do not interfere with the other (e.g., requiring Multi-Factor Authentication for an accounting system might not endanger safety, even though the data is related to an industry where there are several safety-critical systems);
- **Reinforcement:** Properties from both disciplines share the same failure or attack paths and affect the same components (e.g. an attacker can disable a car brake system through an intrusion or the brake system can fail internally, resulting in the same type of accident);
- **Mutual exclusion:** Properties from one discipline negatively affect the properties from another discipline (e.g. adding cryptography to a network connection will add security but will increase latency and that can affect a safety property);

3. Conducting the systematic literature review

The main goal of this SLR is to identify safety and security co-analysis methodologies using model-driven and model-based approaches and how they deal with the interplay between those disciplines, the domains they were applied in both academia and industry, and their tooling support. Having this goal, this SLR was planned, conducted, and reported following well-known review guidelines found in [Kitchenham et al. \(2015\)](#) and [Petersen et al. \(2015\)](#). The SLR planning and conduction are detailed in the following.

3.1. Planning

This SLR addresses five Research Questions (RQ), which are presented as follows, along with the rationale and metrics for each RQ.

- **RQ1.** Which studies propose model-based (MB) or model-driven (MD) approaches to address safety and security concerns in the system engineering process?

Rationale: Understanding the existing model-based and model-driven methodologies that address the co-analysis of safety and security concerns in systems is crucial for advancing the field. This knowledge helps to identify the strengths and limitations of current approaches, fostering the development of new methodologies and tools, and it provides the foundations for future research. Insights into the nature of these studies (i.e., whether they consist in an approach, case study, comparative study, framework, language, metamodel, method, or a pattern) are essential for guiding research efforts and practical applications.

Metrics:

- The number of safety and security co-analysis methodologies using models (MB approaches) and treating models as primary artifacts (MD approaches).
- The number of safety and security co-analysis methodologies categorized by year and by type of contribution (approach, case study, comparative study, framework, language, metamodel, method, pattern).

- **RQ2.** For which domains, life cycle phases, and risk assessment activities have safety and security co-analysis methodologies been proposed and applied?

Rationale: Identifying the domains (e.g., automotive, industrial, medical, transportation), life cycle phases (i.e., requirements, design, development, test, operation), and risk assessment activities (i.e., identification, analysis, control) where safety and security co-analysis methodologies have been proposed and applied is vital for understanding their practical relevance and applicability. Additionally, it is important to identify if industry standards are highlighted. This information provides valuable insights into the effectiveness of these methodologies in various contexts, life cycle phases of system development, and risk assessment. For the research community, it highlights areas of success and gaps that need further exploration, thus guiding future research and development efforts in creating more robust and versatile safety and security co-analysis methodologies and tools.

Metrics:

- Number of types of systems and application domains where safety and security co-analysis methodologies have been applied.
- Number of cited industry standards.
- The list of supported system development life cycle phases.
- The supported risk assessment activities.

- **RQ3.** How do the identified studies address safety and security?
Rationale: Understanding the methodologies employed in the studies is important to comprehend how safety and security are addressed. This involves determining whether safety and security are analyzed together from the start (unified co-analysis) or separately and then combined (integrated co-analysis). Additionally, it is important to identify if the studies aim for co-engineering or cross-fertilization (where one discipline influences the other) and whether the methodologies are quantitative, qualitative, or both. The research community can use this information to evaluate the robustness and applicability of these methodologies and identify areas where further research is needed.
Metrics: Number of studies in each category, presented as following, and relationships among these categories.

- Number of studies per methodology type (unified vs. integrated).
- Number of studies per methodology goal (co-engineering, security-informed safety, or safety-informed security).
- Number of studies by measurement (quantitative, qualitative, or both).

- **RQ4.** How was the methodology proposed by the studies validated?
Rationale: Understanding the validation methods employed by the studies is crucial for assessing the reliability and applicability of the proposed methodologies. This includes identifying whether validation was conducted through comparative studies, empirical studies, expert evaluations, or other methods. Such insights are important for the research community to assess the robustness of these methodologies and to identify best practices for future research and development.
Metrics:

- Number and types of studies related to the validation of safety and security co-analysis methodologies, models, and tools.
- Type of research methods used for validation (comparative study, empirical study, expert evaluation, among others).

- **RQ5.** What are the tools used to support the proposed studies?
Rationale: Identifying the tools used to support the methodologies proposed by the studies helps in understanding their practical application and effectiveness. It is important to determine how each tool was utilized (actively used or merely cited) and how these tools contribute to addressing safety and security disciplines. This information is valuable for the academic community to assess the utility of various tools, identify gaps in tool support, and guide the development of more comprehensive tools for future research and applications.
Metrics:

- List of tools cited or used by the studies.
- Evaluation of how these tools address safety and security disciplines (do not address, address one, or address both).

From the analysis of the research questions, we identified the following relevant keywords: safety analysis, security analysis, model-driven engineering, and model-based engineering. We combined these terms using logical operators (AND, OR) to obtain the following search string.

("safety analysis" AND "security analysis" AND ("model-driven engineering" OR "model-based engineering"))

We avoided using abbreviations like “MDE” for model-driven engineering or “MBE” for model-based engineering. Since the abbreviation is not supposed to appear without its meaning, all papers retrieved

with “model-driven engineering” will also be retrieved with “MDE”, but the opposite is not true. Moreover, in this study we focused on identifying safety and security co-analysis methodologies using modeling approaches. However, the term “analysis” is not adopted unanimously in the literature. We mitigated this problem by conducting a manual search step described in Section 3.2.

We aim to find studies that address safety and security analysis by employing a MB or MD approach. However, it is important to notice that not all authors use these terms in the same way, thus we have to provide our analysis before categorizing the study so that the results respect the definitions we have provided in Section 2.1. We applied this search string to identify studies in the following databases: ACM Digital Library, IEEE Xplore, ScienceDirect, Scopus, and SpringerLink. This search string had to be adapted for each database.

We also defined the inclusion criteria (IC) to identify relevant studies from each database and exclusion criteria (EC) to exclude studies that do not add relevant knowledge to this SLR. A study must meet at least one IC to be included, and it is excluded if it meets at least one EC. The inclusion and exclusion criteria used were:

- IC₁: The study reports a methodology based on or driven by models to support the analysis of the interplay between safety and security concerns.
- IC₂: The study proposes tool support for safety and security co-analysis.
- IC₃: The study presents a case study describing the usage of a methodology or tool to perform safety and security analysis in an application domain.
- EC₁: The study does not propose contributions based on or driven by models.
- EC₂: The study does not propose a methodology or tool for safety or security co-analysis.
- EC₃: The study does not provide enough detailed information about the methodology applied, which makes it impossible to apply the Data Extraction Form.
- EC₄: The study was written in another language than English.
- EC₅: The study addresses (proposes or uses) a methodology or tool that addresses both safety and security, but another more complete study related to the same methodology exists.
- EC₆: The full text of the study was not available through the author’s institutional access, and there was no response from the authors contacted via email.
- EC₇: The study is not peer-reviewed or only available as grey literature (Kamei et al., 2021).
- EC₈: The study must be published in the last 12 years.
- EC₉: The study must be at least four pages long.

To answer the RQs, we defined a Data Extraction Form to extract the following information from the identified studies, presented in Table 1.

3.2. Conduction

We conducted this SLR in two periods: firstly, from July 2022 to December 2022, and then from November 2023 to December 2023, when we updated it as explained in the following paragraphs. We selected and evaluated studies through the procedure presented in Fig. 1. After adapting the defined search string for each database, we obtained a total of 3.319 studies, and after removing duplicates, 3.146 studies remained.

All three authors selected the studies. At least two authors read each study. When there was no agreement between the first two authors, the third author resolved any potential misinterpretations. In cases where the third author’s reading was necessary, a discussion took place to decide whether the study should be included or not, following the defined criteria.

We then conducted the first selection step, where we read all studies’ titles, abstracts, and keywords and applied the selection criteria.

Table 1

Questions of the Data Extraction Form related to the research questions they aim to answer.

RQs	Fs	Questions
RQ1	F ₁	What is the contribution of the study? The possible values are approach, case study, comparative study, framework, language, method, and pattern.
RQ1	F ₂	Does the study propose or use any MB or MD methodology to co-analysis of safety and security? If so, describe the proposed contribution with advantages and disadvantages.
RQ1	F ₃	Is there a case study described? If so, describe it.
RQ2	F ₄	What is the application domain targeted by the study?
RQ2	F ₅	Which are the software life-cycle phases that the study addresses?
RQ2	F ₆	Which are the risk assessment activities addressed by the study?
RQ2	F ₇	Is there any industry standard highlighted in the study? If so, what are they?
RQ3	F ₈	Does the study successfully address safety and security co-analysis? If so, specify which process the study adopts. • Safety and security were analyzed separately in different models and put together later in development (Integrated); • Safety and security were analyzed on the same model from the very beginning (Unified);
RQ3	F ₉	What is the analysis goal? • To accomplish simultaneous analysis in a twofold objective (Co-engineering); • To accomplish cross-fertilization (Safety-informed security or Security-informed safety).
RQ3	F ₁₀	Is the study's approach quantitative, qualitative, or both?
RQ4	F ₁₁	How was the methodology evaluated? • Comparison with other methodologies (Comparative); • Execution of the analysis on a controlled environment (Empirical); • Expert evaluation; • Application to a case study;
RQ5	F ₁₂	Does the study use any tool to support the analysis? If so, specify what disciplines the tool addresses with native features and if the tool was used or just cited.

Following the second selection step, we read the full text of all studies and applied the selection criteria again. The first selection reduced the number of primary studies to 134, and the second selection reduced it to 35 primary studies.

After conducting the first and the second selection steps, aiming to identify the primary studies, we conducted a backward snowballing inspection in each selected study. During the snowballing, we performed the first and second selection steps to analyze the references for each primary study. We repeated this process until no more primary studies were found. At the end of the snowballing inspection, we included 40 other relevant studies, resulting in 75 primary studies.

During this SLR, we faced different terms in the academic community that had similar meanings. For example, the paper from [Dobaj et al. \(2019\)](#) uses the word “co-design” in its content and “integrated” in its title because they apply safety and security co-analysis during the design phase of the system, but they do not use the terms “co-engineering” or “co-analyses”. The paper from [Martinez et al. \(2020\)](#) used the word “interference” to show how safety and security can be analyzed together. This divergence of terms led us to conduct an additional manual search step (right side from [Fig. 1](#)), which resulted in 21 new studies.

The manual search involved adding 2 studies recommended by specialists and 19 studies identified using an alternative search string, totaling 21 studies. The revised search string aimed to address the issue of term divergence discussed in the previous paragraph by employing a broader approach. This updated strategy was subsequently applied in the SLR protocol. We searched the same 5 databases (refer to [Section 3.1](#)) and applied inclusion and exclusion criteria to all studies based on their titles, abstracts, and keywords. These criteria were then reapplied to the full texts of the selected studies.

("security" AND "safety") AND ("co-analysis" OR "co-engineering") AND ("model-based" OR "model-driven")

We also conducted a second backward snowballing inspection on these 21 new studies resulting from a manual search, as well as secondary studies in this area identified during the analysis of the related work (see [Section 7](#)). We conducted the second snowballing inspection similarly to the first one, repeating it until no more studies were found. We analyzed the references of each relevant study and performed the first and second review selection steps guided by the same inclusion and exclusion criteria. This inspection resulted in 23 other relevant studies, totaling 44 new studies.

We extracted relevant information from each primary study accepted after the first selection, second selection, and snowballing inspection by filling out the Data Extraction Form presented in [Section 3.1](#).

4. Results

We could extract relevant information from the 119 identified studies to answer our RQs successfully. To facilitate the reference to all studies, they received a Study ID, presented in [Appendix A](#). All the extracted data is located in a table elsewhere² due to space restrictions. In the following sections, we will present the absolute number of studies as an integer, followed by a percentage with two decimal places. The percentage is an approximation to help readers quickly understand the participation of that categorized group of studies among all identified studies.

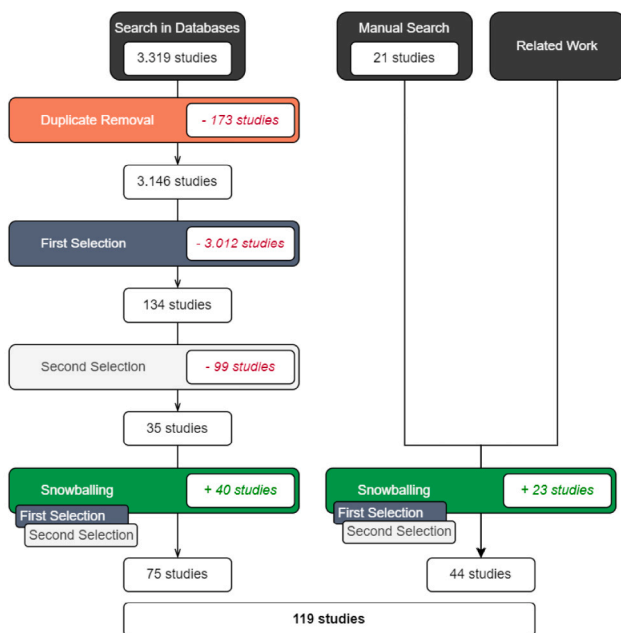


Fig. 1. Overview of our Conducting phase.

² <https://bit.ly/3xJCHEI>

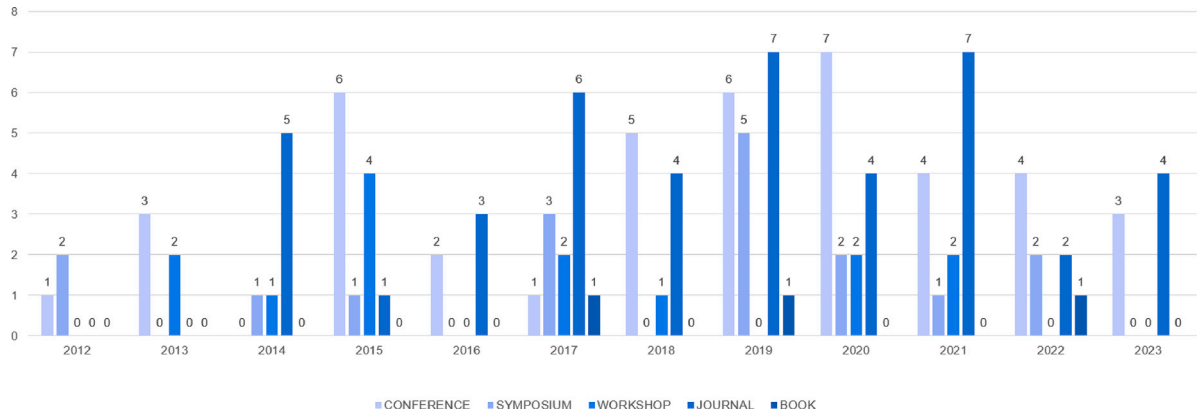


Fig. 2. Number of studies grouped by their publication year.

Table 2

List of studies grouped by their contribution and modeling method.

Study ID	Contribution	Modeling method
23, 31, 34, 38, 39, 46, 54, 58, 61, 65, 70, 72, 73, 74, 87, 88, 90, 91, 93, 94, 98, 99, 104, 106, 109, 118	Approach	MB
5, 10, 25, 29, 49, 51, 53, 66, 68, 75, 82, 100, 101, 103, 113, 114	Approach	MD
47, 55, 56, 79	Case Study	MB
9, 57, 83	Case Study	MD
2, 20, 36, 102, 105	Comparative Study	MB
43, 60, 76, 78, 92, 97, 108, 116	Framework	MB
17, 89, 96, 119	Framework	MD
50	Language	MB
52	Language	MD
77	Metamodel	MB
4, 13, 14, 19, 21, 27, 28, 30, 32, 33, 35, 37, 41, 42, 44, 59, 62, 63, 64, 67, 69, 71, 81, 85, 86, 102, 111, 112, 115, 117	Method	MD
1, 6, 7, 8, 11, 12, 15, 16, 18, 22, 24, 26, 40, 45, 48, 80, 84, 107, 110	Method	MD
3, 95	Pattern	MB

4.1. Answers to RQ1

We present the number of studies grouped by their publication year and type of publisher as a timeline in Fig. 2. More complete information regarding the publication of those studies is presented in Appendix B. The subject of those publishers differs considerably. We have publishers with specific subjects related to this work (e.g., “safety and security”, “security and privacy”, or “dependability”), but we also have middle-terms (e.g., “computer science in general”) and subjects in general.

In Table 2, each line groups similar studies based on their contribution to the literature and modeling approach.

Regarding the methodology, the difference between method and approach is worth mentioning. A method is a defined sequence of reproducible steps, while an approach is a different point of view. An approach can be created over a technique (e.g., FTAs and its analog version AT), but also over a method (e.g., STPA and its extension using STRIDE, proposed by de Souza et al. (2020), where STPA is extended by adopting STRIDE in one of its four steps, the *Identify Loss Scenarios* step). To summarize, a new approach to a method can be a change on only a few steps of the current method, without the need to define an entirely new method. From our extraction, 48 primary studies (40,34%) contributed with a method, 42 primary studies (35,29%) contributed with an approach.

We stick with primary studies authors’ definitions whenever possible for other contributions. Only 12 primary studies reported a framework, 2 reported a language, 2 reported a pattern, and 1 reported a metamodel. The work of Mashkoor et al. (2022) explains those types of contributions. Framework is a platform providing a foundation for safety and security systems development. Languages can be formal languages, graphical modeling languages, architecture description language, programming languages, or language for presenting mishaps.

We also found 7 primary studies reporting a case study, where studies use a method, approach, or technique defined in other works and apply it to a case study. Finally, 5 comparative studies were found where authors compare two or more methods, approaches, or techniques.

For the modeling approaches, as defined in Section 2.1, we identified 76 primary studies (63,87%) of MB methodologies against 43 primary studies (36,13%) of MD methodologies. MDE is strongly supported by models that are formally represented. According to Brambilla et al. (2012), to make model transformations or generate code, the meta-model (of those models used in the identified methodologies) must be written in a formal language with semantics defining concepts, properties, relationships, and constraints. As mentioned in Section 2.4, formal methods are outside the scope of this SLR. Therefore, they are absent in the results.

4.2. Answers to RQ2

Extensive diversity of application domains was identified by this SLR, as can be seen in Fig. 3. This reflects the broad applicability of safety and security disciplines in software engineering. The most present domain is transportation, with 57 primary studies (47,89%) of presence, divided into automotive with 34 primary studies (28,57%), autonomous vehicles with 11 primary studies (9,24%), avionics with 7 primary studies (5,88%), railroad and ships with 2 primary studies (1,68%), and autonomous drones with 1 primary study (0,84%). The automotive domain is the most expressive due to the introduction of the ISO 26262 (ISO, 2011) international standard for functional safety in passenger vehicles, which has greatly increased the awareness within the industry for the issues involved in functional safety (Burton et al., 2012). From 34 primary studies addressing the automotive domain, 21 of them cited the standard. In the second place, we identified the industrial domain with 25 primary studies (21,01%).

We used the definition of Sommerville (2015) for the life cycle phases of system development. Those phases are: *Requirements* for the specification of objectives, services, and restrictions of the system; *Design* for the design of system functionality and architecture; *Implementation* for the development of the system; *Testing* for the validation and verification of the system; and, *Operation* for the usage and maintenance of the system.

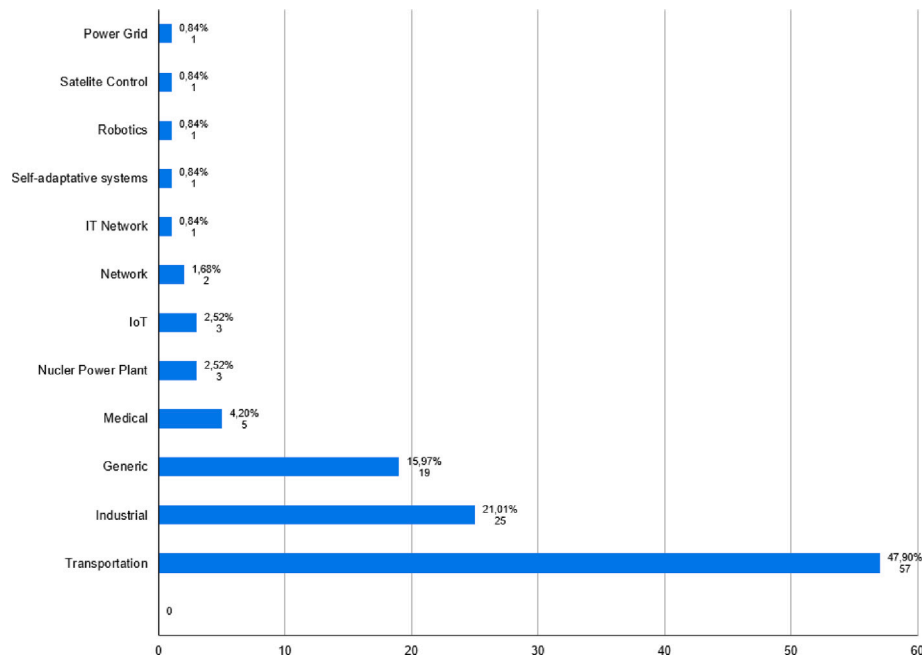


Fig. 3. Number of studies grouped by their application domain.

Table 3

List of studies grouped by the system development life cycle phases.

Study ID	Life cycle phase				
	Requirements	Design	Implementation	Testing	Operation
78, 97	✓	✓			✓
2, 3, 4, 10, 11, 14, 45, 48, 51, 55, 80, 83, 90, 100, 113, 119	✓	✓			
77		✓			✓
1, 5, 6, 7, 8, 12, 13, 20, 22, 25, 26, 29, 32, 33, 34, 35, 36, 39, 40, 41, 42, 43, 44, 47, 49, 50, 52, 53, 54, 56, 57, 58, 59, 60, 62, 63, 64, 65, 66, 67, 68, 69, 71, 73, 74, 81, 82, 85, 87, 88, 92, 93, 96, 99, 101, 102, 103, 104, 105, 106, 107, 109, 110, 111, 112, 115, 117		✓			
9, 15, 16, 17, 18, 19, 21, 24, 27, 28, 30, 31, 37, 38, 46, 61, 70, 72, 75, 76, 84, 86, 91, 94, 95, 98, 114, 116, 118	✓				
23, 89, 108				✓	✓
79				✓	

When analyzing the application domain of the studies to the known life cycle phases of system development, we identified that most of the studies address the requirements or design phase. We identified these life cycle phases for every primary study during our extraction. Also, only 19 primary studies (15,97%) addressed more than one phase. In Table 3, we present the studies grouped by those phases.

We can notice in Table 3 that the Implementation life cycle phase is not addressed at all. Almost all studies focused their methodologies on the Requirements or Design phase. Only 6 studies explicitly consider the applicability of their methods when the system is already operating (O), and only 1 study addresses the Test life cycle phase.

Table 4

List of studies grouped by the risk activities addressed.

Study ID	Risk activity		
	Identification	Analysis	Control
1, 3, 5, 9, 13, 16, 19, 21, 24, 33, 35, 37, 42, 45, 54, 67, 72, 73, 77, 78, 87, 88, 90, 102, 105, 107, 108, 109, 111, 115, 119	✓	✓	✓
2, 4, 18, 20, 23, 27, 28, 31, 36, 39, 46, 47, 48, 51, 55, 56, 58, 59, 60, 63, 68, 70, 71, 75, 81, 83, 86, 92, 95, 98, 100, 104, 106, 112, 117, 118	✓	✓	
12, 22, 40, 43, 62, 101		✓	✓
97	✓		✓
6, 10, 11, 14, 15, 25, 26, 38, 41, 44, 50, 52, 53, 57, 61, 65, 66, 69, 7476, 80, 82, 93, 94, 96, 103, 110, 113, 116		✓	
7, 8, 17, 29, 30, 32, 34, 49, 79, 84, 89, 91, 99, 114	✓		
85			✓

We considered identification, analysis, and control activities for risk assessment activity. As the name indicates, identification is the activity where hazards and threats are identified. Analysis assesses the risk itself and considers several parameters, such as severity, likelihood, impact, cost, and damage. The control activity is the suggestion of countermeasures to reduce risks. Those activities are sequential, but only 32 primary studies (26,89%) addressed all three activities. Previous activities are considered a preliminary step for applying the methodology for the remaining studies.

The studies grouped by the Risk Assessment Activity are presented in Table 4, the Analysis activity is the most present, being addressed by 103 (86,55%) studies, followed by Identification with 83 studies (69,75%) and Control with 40 studies (33,61%).

4.3. Answers to RQ3

To answer RQ3, three different charts are presented. First, in Fig. 4, we present the number of studies according to their methodology type.

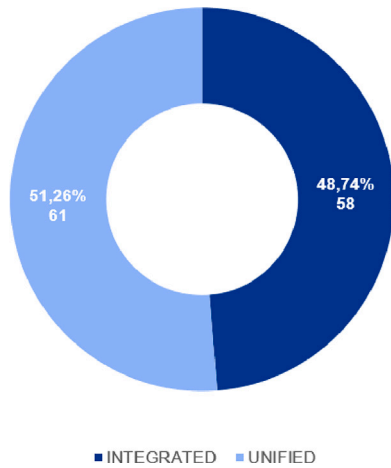


Fig. 4. Percentage of studies grouped by their methodology type.

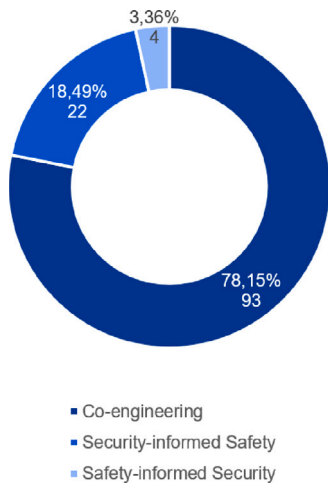


Fig. 5. Percentage of studies grouped by their methodology goal.

The possible values are (i) *Integrated* when safety and security analysis are conducted separately but at some point, their artifacts have to be merged and submitted through a trade-off analysis (Nigam et al., 2018) or harmonization process (Nicklas et al., 2016); and (ii) *Unified*, when safety and security analysis are conducted closely, requiring exchange and discussion between both disciplines in all phases of the system engineering life cycle (Schmittner et al., 2015). We identified an equilibrium between those methodology types.

The second chart, Fig. 5, presents the primary studies classified by their methodology goal. Co-engineering (simultaneous analysis), security-informed safety (safety analysis improved with security concerns), and safety-informed security (security analysis improved with safety concerns) are the possible values. Co-engineering is the most present goal in our SLR with 93 studies (78,15%), followed by security-informed safety with 22 studies (18,49%). This is due to the maturity of safety analysis over security analysis, as mentioned in Sections 1, 2.2 and 2.3. Indeed, it is easier to modify a well-established safety analysis to consider security concepts than the opposite.

Lastly, Fig. 6 presents the primary studies classified by their measurement, in which the possible values are quantitative, qualitative, or both. Quantitative measurement is considered when the methodology works with failures and threats' likelihood, probability, Mean Time To Failure, risk, or other parameters as real numbers. Methodologies that classify those parameters by range (e.g., failure impact classified as low, medium, or high) are considered qualitative. Quantitative methodologies are less present in the literature, with only 12 primary studies

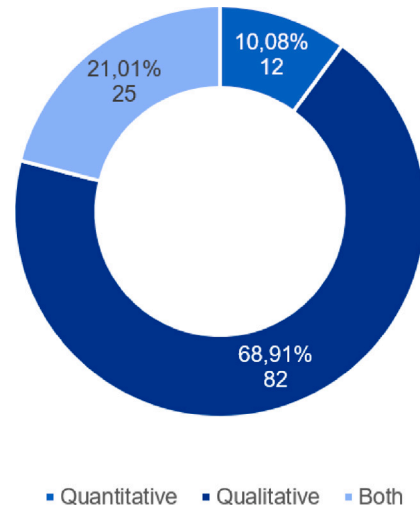


Fig. 6. Percentage of studies grouped by their measurement.

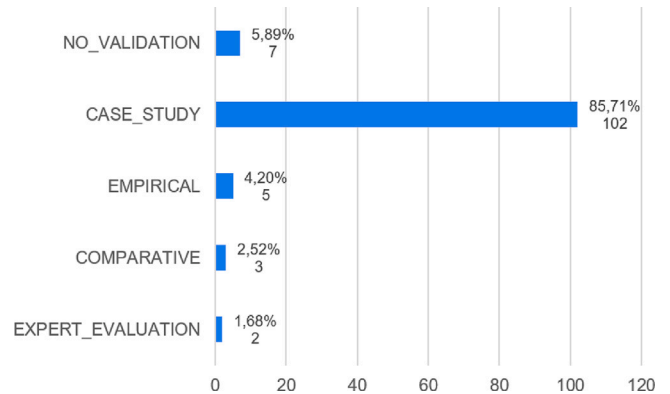


Fig. 7. Number of studies grouped by their evaluation method.

(10,08%) plus other 25 primary studies (21,01%) considering the *Both* classification.

4.4. Answers to RQ4

Most of the identified studies were evaluated by a case study (102 primary studies representing 85,71%) that applies the proposed methodology, as seen in Fig. 7. Those case studies vary from hypothetical and simplified systems to real systems. Only 5 studies were evaluated in empirical experiments (4,2%), 3 are comparative studies (2,52%), and the other 2 are expert opinions (1,68%). The last 7 (5,89%) primary studies did not provide any validation.

4.5. Answers to RQ5

There were 93 mentions of tools that support the proposed methodology, concentrated in 52 (43,7%) of the identified primary studies. The most mentioned tools are Rodin, with 7 mentions, and TTool, with 6 mentions. There were 53 different tools mentioned in primary studies. Some studies mention up to 8 tools, while the majority mention only one, representing various tools in the literature. Besides, some studies do not use the tools they present; instead, they present their potential. The categorization according to the type of mention helps make that distinction. From 93 mentions, in 69 (74,19%) tools were used, while in 24 (25,81%) tools were just cited.

Regarding the discipline supported by the identified tools, 19 tools (35,85%) do not address safety or security. They were used for modeling purposes such as behavior models, system design, and data flows. Tools that address both disciplines as native features represent 37,74% (20 tools). Tools that address only safety or only security represent, respectively, 15,09% (8 tools) and 11,32% (6 tools). Table 5 presents all the tools cited in primary studies.

5. Discussion

In this section, we discuss several aspects that emerged from the analysis of the SLR results.

5.1. Process-based versus component-based methodologies

After observing the results of the 119 selected studies, two groups of methodologies were identified: process-based and component-based. In the context of this work, process-based methodologies consider the data flow and how the system interacts with the physical environment.

The following process-based methodologies were identified in this SLR: (i) STAMP S&S (Kaneko et al., 2020; Kaneko and Yoshioka, 2020); (ii) STPA-SafeSec (Friedberg et al., 2017); (iii) Systems-Theoretic Likelihood and Severity Analysis (Temple et al., 2017); and (iv) Uncontrolled Flows of Information and Energy (Carreras Guzman et al., 2021a) as process-based methodologies.

In the same way, component-based methodologies focus on smaller pieces of the system, and the results can be merged for an artifact representing the entire system. The Attack Tree BowTie (Abdo et al., 2018), Attack-Fault Trees (Kumar and Stoelinga, 2017), Parametric weighted timed automata (Andre et al., 2019), Pattern Engineering Lifecycle (Amorim et al., 2017), and Failure Mode, Vulnerabilities and Effect Analysis (FMVEA) (Schmittner et al., 2014b) are some of those methodologies.

5.2. Modeling approaches

As presented in Section 4.1, we identified 76 primary studies of MB methodologies against 43 MD methodologies. This represents 76,74% more studies on MBE than in MDE. Another fact we have detected is that primary studies proposing MD methodologies addressed less risk assessment activities (RAA) than MB methodologies, as presented in Table 6. A possible justification for this is the effort needed to develop meta-models, implement changes to existing modeling tools, or even develop a brand-new tool to support the proposed meta-model and methodology. Only 25 tools were cited along with MB methodologies, against 68 tools cited by MD methodologies. More about tooling is presented in Section 5.6.

MDE proposes the reuse of modeling artifacts for code generation and model transformations. This facilitates the achievement of lower development costs in the long term by increasing the efficiency of engineers in safety and security analysis. It is worth mentioning that the initial effort, cost, and learning curve can be higher when applying MDE, and it can configure itself as an obstacle to be overcome.

5.3. Application domain

Safety and security are more applicable to critical systems, such as transportation and industrial domains. The answer to our RQ2 (presented in Section 4.2) showed the majority presence of those domains in primary studies. That does not mean exclusivity since 37 primary studies (31,09%) addressed other domains. Safety and security concerns are becoming widely used, but there is still room for addressing other application domains other than transportation and industrial.

We also identified a strong relationship between industry standards and those domains. Supporting the Automotive domain, there is the already mentioned ISO 26262 (presented in Section 4.2 regarding

Table 5

List of tools, the disciplines addressed, and how many times they were mentioned by the primary studies. The checkmark represents that the discipline is addressed. Mentions are categorized between *Used by* and *Cited by*.

Tool name	Safety	Security	Used by	Cited by
Alloy analyzer	✓	✓	1	0
AltaRica			1	1
Architecture analysis & Design Language (AADL) Error Model V2	✓		0	1
ATTop Tool	✓	✓	1	1
Breach			1	0
Cameo systems modeler			1	0
Capella			1	2
CHESS	✓	✓	2	0
CLINGO			1	0
CMSS	✓	✓	1	0
Concept-aware analysis library			1	0
Cyber architect		✓	2	0
Cyber security modeling language (CySeMoL)		✓	0	1
Enterprise architect 14			1	0
EPF-C ³ - (Eclipse Process Framework - Composer)			1	0
ERIS			1	0
ETSI	✓	✓	1	0
Figaro			1	0
Figseq	✓	✓	2	0
FIIS tool	✓	✓	1	0
FinPSA	✓		1	0
Henshin			1	0
KB3 workbench			2	0
Labelled transition system analyser (LTSA)			1	0
MATLAB			3	0
Melody			1	0
Microsoft Threat Modeling		✓	2	0
MILS-AADL	✓	✓	1	0
MSBNx			1	0
Multihost, multistage Vulnerability Analysis (MulVAL)		✓	0	1
OCRA	✓		1	0
Open Source AADL Tool Environment (OSATE)	✓	✓	1	0
Papyrus			2	2
Papyrus-RT			0	1
PolarSys OpenCert	✓	✓	2	0
Proposed tool SOT	✓	✓	1	0
ProTL	✓	✓	1	0
ProVerif		✓	1	4
REAFFIRM	✓		1	0
Rodin	✓		4	3
Safety Architect	✓	✓	3	0
Simulink			2	1
STAMP Workbench	✓		1	0
ThreatGet Plugin		✓	1	0
ToSS (Traceability of Safety Systems)	✓	✓	1	0
TTool	✓	✓	5	1
UPPAAL	✓		2	3
Uppaal SMC	✓		2	0
Vitech GENESYS MBSE	✓	✓	1	0
WEFACT (Workflow Engine for Analysis, Certification and Test)	✓	✓	1	0
XSTAMPP (eXtensible STAMP Platform)	✓	✓	1	0
YAKINDU Security Analyst (YSA)	✓	✓	1	0
YAMS	✓	✓	0	2

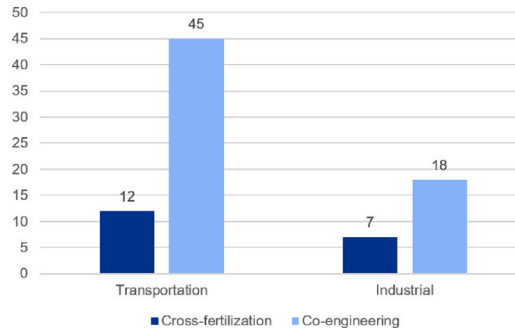


Fig. 8. Number of studies grouped by methodology goal and application domain. Only industrial and transportation application domains are presented.

Table 6

Number of studies by their modeling approach and number of RAAs addressed.

	Number of RAAs addressed			Total
	Only one	At least two	All three	
MB	20	32	24	76
MD	24	11	8	43
Total	44	43	32	119

automotive safety) and the ISO/SAE 21434 (ISO, 2021), which focuses on automotive cybersecurity engineering. With this support, academic researchers can develop more methodologies by adapting or improving those standards to new situations in this domain. The Industrial domain also has support from the IEC 61511 (Functional safety - Safety instrumented systems for the process industry sector) (IEC, 2016). Thus, academic researchers can enjoy the same benefits for safety in the industry sector, but not for security. Other domains do not have these specific standards to support. Instead, they have been using more generic standards like IEC 61508 (Functional safety of electrical/electronic/programmable electronic safety-related systems) (IEC, 2010), but there is also a lack of security concerns for those domains.

Although we identified that transportation and industrial sectors are supported by industry standards, which are concerned with a single domain, when put in context with the methodology goal used, we can see that academic research is biased towards the co-engineering goal. The industrial application domain aiming at co-engineering is present in 18 studies, against 7 studies aiming at cross-fertilization. The transportation application domain has 45 studies applying co-engineering against 12 applying cross-fertilization. We highlight this bias in Fig. 8. A more complete explanation of methodology goals is presented in Section 5.5.

Medical and Internet of Things (IoT) are application domains that appeared only a few times during this SLR, with 5 and 3 studies, respectively. Nevertheless, they could be identified together on the same application, for example, in the PCA Interlock application described by Procter et al. (2017). Medical devices were originally designed to operate in standalone environments, but new features have been conceived by connecting devices to the Internet (Hayakawa et al., 2018). Since IoT connects *things* to the *Internet*, it can intersect with any other domains, increasing their exposure to cybersecurity threats. Therefore, we expect an increase in the participation of other domains when conducting future literature searches on safety and security analysis.

5.4. System development and risk assessment support

As presented in Section 4.2, the life cycle phases of system development most addressed are *Requirements* and *Design*. The *Implementation*, *Testing*, and *Operation* life cycle phases were addressed by 0, 1, and 6 studies, respectively. This distribution is expected since we focused on model-driven methodologies, and models are usually artifacts built during the initial phases of system development. Our focus on safety

Table 7

Number of studies by their methodology goal and methodology type.

	Integrated	Unified
Co-Engineering	32	61
Security-informed Safety	22	0
Safety-informed Security	4	0

and security also reinforces this result. During the co-analysis of these disciplines, we expect to interact and analyze their interplay at early stages to avoid waste of effort and higher costs. More about the interplay of safety and security is present in the next section.

Regarding risk assessment activities, most of the studies addressed the *Analysis* activity, which is also expected due to our focus on safety and security co-analysis. The *Identification* activity has a strong presence, too, but it does not overcome the *Analysis* activity. As mentioned in Section 4.2, some methodologies considered the *Identification* as a preliminary activity for applying the *Analysis* activity. Thus, the *Identification* is not addressed in the study. The *Control* is the least present activity of all. Only 1 study addressed it in isolation, considering *Identification* and *Analysis* as preliminary activities. Methodologies that address *Control* and other previous activities are more robust and may be more challenging for the research community to produce.

5.5. Interplay between safety and security

We identified an equilibrium between *Integrated* and *Unified* methodologies (see Section 4.3). When considering the publication year in the analysis, we noticed a balanced cadence between studies proposing integrated and unified approaches, as presented in Fig. 9. This shows that academia has not yet established a preferred methodology.

Another relevant analysis can be conducted by crossing the study's methodology goal with the methodology type. The *Co-Engineering* goal was more found along with the *Unified* methodology type, represented by 61 primary studies (51,26%), but it also appears in the *Integrated* methodology type, represented by 32 primary studies (26,89%). Cross-fertilization goals, named *Security-informed Safety* and *Safety-informed Security*, are only found in integrated approaches, with 22 primary studies (18,49%) and 4 primary studies (3,36%), respectively. Table 7 presents those relations.

As mentioned in Section 4.3, integrated approaches that aim to accomplish co-engineering must have a step for trade-off analysis or harmonization since safety and security disciplines can mutually exclude one another. This additional step can only occur at interaction points during the process (Pomante et al., 2019) or at the end of the development process, which may bring unwanted surprises in effort and cost. According to Lisova et al. (2019), integrated approaches that consider safety as an overall goal also have a potential limitation when applied to systems where non-safety-related security issues are important. In such a case, security analysis would have to be performed separately. The required effort can be reduced during the harmonization step, but there would still be duplication of work. Similar

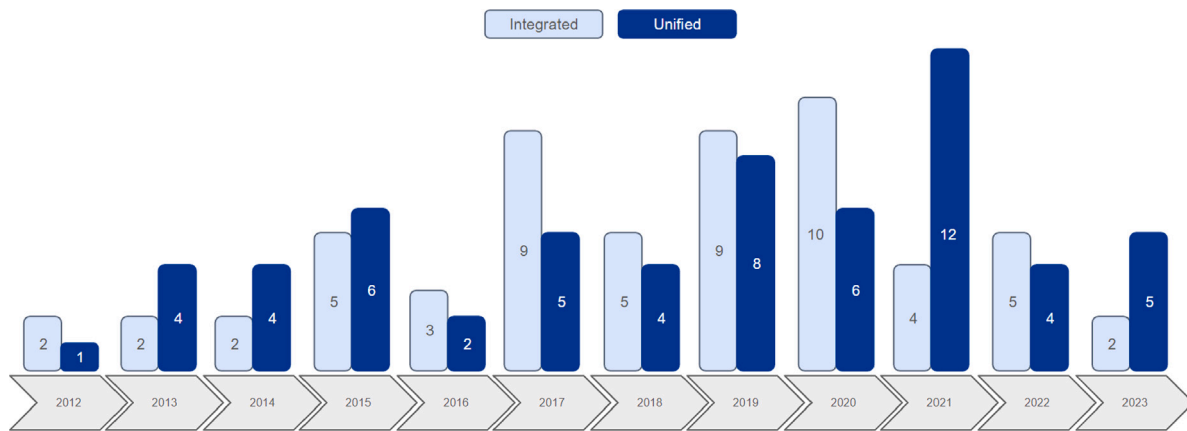


Fig. 9. Number of studies grouped by methodology type and publication year.

activities will occur in unified approaches, but since both disciplines are conducted closely, the effort is better distributed along the process. The identified studies that presented a unified methodology type did not create a step for trade-off analysis or harmonization because they did not have to. This essence of unified approaches is an advantage compared to integrated approaches since we can avoid unwanted surprises. Still, it also comes with the disadvantage of combining different profiles of engineers, which may lead to stressful teamwork.

Regarding the maturity of safety and security disciplines, the first study that proposed an approach with a *Safety-informed Security* goal was published in 2019, while *Co-Engineering* and *Security-informed Safety* first studies were published in 2012, respectively. Therefore, the most common strategies start with a safety analysis and increment it with security concerns or consider both from the beginning. Security analysis will likely achieve better maturity and become another possible starting point for assessing critical systems in the following years.

This SLR has identified several distinct approaches, methods, techniques, and processes, but another relevant aspect of safety and security co-analysis is worth mentioning: community acceptance. Since safety and security have been treated separately (see Sections 1 and 2.5), security and safety analysts also work individually and know too little about the other discipline. The work of Kruck et al. (2021) is among the identified studies and tackles this issue by proposing a MB workflow that allows safety and security analysts to learn about propagation considerations of the respective other discipline. The workflow was created over a metamodel and gives suggestions to analysts instead of running an analysis at the tool's backend and outputting a result that might not need to be clarified to all analysts.

5.6. Tooling

As presented in Section 4.5, the majority of tools support both disciplines with native features, such as TTool, which is based on the SysML-Sec methodology (Zhao et al., 2019); CMSS, a prototype tool created by the authors of the study (Liu et al., 2021); STAMP-based Ontology Tool (SOT), also created by authors for this purpose (Pereira et al., 2019); and PolarSys OpenCert.³ (Javed et al., 2021) This identified majority highlights the importance of tools that approximate safety and security disciplines when conducting their co-analysis.

Other tools accomplish this feature by developing plugins or extensions, such as CHESSE,⁴ improved with SafeConcert (Gallina and Haider, 2020); Safety Architect,⁵ which supports security with an extension (Brunel et al., 2014); and Enterprise Architect 14⁶ with ThreatGet Plugin.⁷ (Martin et al., 2020)

Some primary studies defined their methodology around tools, which can use general-purpose tools for system design, behavior modeling, data flows, and specific tools for each application domain. Some examples are presented by Kriaa et al. (2015a, 2019), which use KB3 Workbench, Figaro language,⁸ and Figseq in their S-Cube approach; and by Martinez et al. (2020), which uses Safety Architect, Cyber Architect,⁹ and Concept-aware analysis library in their interference analysis. A very present combination in primary studies uses a modeling tool that supports dependability properties (e.g., TTool and CMSS), along with UPPAAL and ProVerif as backend verification tools for safety and security, respectively.

Some studies accomplish safety and security co-analysis by using attacks as some kind of *external failure*. Therefore, every tool that can be used for safety now also supports security. This approach may work but has a conceptual limitation. As mentioned in Section 2.3, an attack consists of expertise, time, knowledge, access, and resources needed to make an attack feasible. Other properties, such as vulnerabilities, can also integrate the concept of an attack and can be decomposed into several properties on their own. Using this approach may put these relevant properties aside. This conceptual limitation can be overcome with more meta-models that address safety and security in completeness.

6. Threats to validity

We are aware of a few threats that could impair the validity of our study.

Construct Validity. Our search strategy could not cover all publications regarding safety and security. More than one search string was tested before reaching the final version used in this work. For example, using the term *cybersecurity* instead of *security* could result in more specific primary studies but could also neglect other works similar in

³ <https://www.eclipse.org/opencert/>

⁴ <https://www.eclipse.org/chess/>

⁵ <https://www.all4tec.com/en/safety-architect-fmecca-fta-software/>

⁶ <https://sparxsystems.com/products/ea/14/index.html>

⁷ <https://documentation.threatget.com/20.09/>

⁸ <https://github.com/charles-river-analytics/figaro>

⁹ <https://www.all4tec.com/en/cyber-architect-en/>

importance for this SLR. To overcome this threat, we conducted an additional manual search using a second search string and a backward snowballing inspection of all selected papers.

External Validity. Some publications described their research in great detail, while others provided only sparse, sometimes even vague details. For example, we identified proposed approaches without validation. At the same time, other authors published more than one work regarding the same methodology applied to different scenarios or passing through different types of validation. This shows a need for more consistency and completeness in the available information, making the data extraction highly dependent on our interpretation. Moreover, our findings are based on modeling approaches to safety and security co-analysis. The generalizability of these conclusions to methodologies that do not use models may be limited. Future research could explore the applicability of our conclusions in non-model-based methodologies to further validate and refine the insights presented in this study. To overcome this threat we applied the exclusion criteria EC₃, which excludes papers that do not provide detailed information, and at least two authors assessed all papers.

Internal Validity. Some authors provided ambiguous information or used concepts in a different way than we did. For example, the publication uses the term *model-based*, but in our understanding, it proposes a model-driven approach. We classified the studies according to our interpretation. This threat was overcome by the same measures described in external validity. When the study did not provide enough detailed information to be part of the results, it was excluded during the selection steps. Also, all papers were assessed by at least two authors, which elevates our interpretations to agreement.

7. Related work

Safety and security co-analysis is a relevant topic presented in several secondary studies, and it can be enhanced by adopting MDE. Those studies were our source of inspiration and drove the classifications used in this current work. Table 8 presents these classifications by secondary study.

7.1. Systematic literature reviews

The work of Lisova et al. (2019) is very similar and was our primary source of inspiration. They researched the most adopted techniques for co-analysis of safety and security but focused on the requirements life-cycle phase of software development. Their results showed a need for more evaluation of the proposed methodologies. This gap was addressed by Pirbhulal et al. (2021), who added a specific research question to understand how the identified techniques in their study were evaluated, and results showed that a more extended investigation is necessary to estimate the efficiency and accuracy of those validation methods. Their work focused only on Graph Theory and Bayesian Network models. The work of Pekaric et al. (2023) is a complete work specific to self-adaptive systems, categorizing hazard sources, cause and handling, and attack mechanisms and surfaces. They analyzed the usage of models regarding safety, security, and architecture.

Our work differs from the previous SLRs on safety and security co-analysis by removing limitations of scope. There is a model limitation in the (Pirbhulal et al., 2021) study, which focuses only on network models. A limitation regarding the system type is present in Pekaric et al. (2023), since they focus on self-adaptive systems. Finally, the (Lisova et al., 2019) study limited its scope to the requirements life-cycle phase.

7.2. Systematic mapping studies

According to Petersen et al. (2015), “While systematic reviews aim at synthesizing evidence, also considering the strength of evidence, systematic maps are primarily concerned with structuring a research area”. We found three systematic mapping studies that also targeted the co-analysis of safety and security. The work of Kenner et al. (2021) focused on configurable systems, and the results are enumerated as literature gaps, which provide clear future directions for scientific production. A mapping study of safety and security analysis methods restricted to autonomous vehicles is presented by Boyer et al. (2021). The study from Mashkoor et al. (2022) is the only mapping study that defined research questions and aimed to map MDE methods for safety and security, as we did. Thus, they presented fewer classifications in the results.

7.3. Comparative studies, reviews and surveys

Other secondary studies extracted relevant information that drove our classifications (Pietre-Cambacedes and Bouissou, 2013; Kriaa et al., 2015b; Paul and Rioux, 2015; Abulamddi, 2016; Chockalingam et al., 2017; Torkildson et al., 2018; Kavallieratos et al., 2020; Lyu et al., 2019; Oueidat et al., 2020). These works are surveys, reviews, and comparative studies. Therefore, they did not follow a systematic protocol like those in SLRs and systematic mapping studies.

A comparison study between FMVEA, CHASSIS, and STPA-Sec techniques applied to an autonomous boat case study is presented by Torkildson et al. (2018). They evaluated the level of abstraction, replicability, reusability, scope of analysis, suitability to risk rating, and adaptability to the context of those techniques. The work of Oueidat et al. (2020) also compared several approaches by applying them to a beer production system. They evaluated if the approach is generic or MB, qualitative or quantitative, level of detail, integration principles between safety and security, software life-cycle phases, and application domains.

With a focus on CPSs, we can highlight the works from Lyu et al. (2019), Kavallieratos et al. (2020), and Abulamddi (2016). Other works focused on safety and security interplay (Kriaa et al., 2015b; Pietre-Cambacedes and Bouissou, 2013; Paul and Rioux, 2015). Finally, the last related work contributed with a survey relating to safety and security integration with risk assessment (Chockalingam et al., 2017).

In summary, we took advantage of the categorization and limitations presented by 15 secondary studies related to safety and security co-analysis to define the planning of our SLR. In this SLR, our focus was on safety and security co-analysis by using modeling approaches. We addressed all the identified aspects without putting any restrictions on application domains or types of systems (e.g., CPS or self-adaptive systems).

8. Conclusion

We performed an SLR to obtain a panorama of the current research on safety and security co-analysis methodologies with the aid of models. We identified 119 primary studies by answering the RQs regarding the interplay between safety and security analysis methodologies, highlighting potential future research directions such as model-driven safety and security methodologies dedicated to formal analysis and verification; case studies addressing safety and security co-analysis in other application domains (e.g., avionics and CPS); secured-informed safety analysis; and the need for more mature tooling support for safety and security co-analysis.

We have observed that although the current state-of-the-art offers practical aid for co-analyzing safety and security using model-driven methodologies, its adoption rate is lower than expected. Specifically, MBE was mentioned in 34 more studies than MDE. The steep learning

Table 8
Related work and the classification used in their methodology.

Study reference	Study type	Classifications									
		Applica- tion domain	Standards	Tool support	Methodol- ogy type	Methodol- ogy goal	Modeling approach	Measure- ment	Software life cycle phase	Risk as- sessment activity	Validation of ap- proaches
Lisova et al. (2019)	SLR				✓	✓					
Pirbhulal et al. (2021)	SLR	✓									✓
Pekarić et al. (2023)	SLR	✓					✓			✓	✓
Mashkoor et al. (2022)	Systematic Mapping	✓							✓		
Kenner et al. (2021)	Systematic Mapping	✓	✓	✓		✓					
Boyer et al. (2021)	Systematic Mapping					✓					
Torkildson et al. (2018)	Comparative Study										
Oueidat et al. (2020)	Comparative Study	✓				✓					
Lyu et al. (2019)	Review					✓	✓	✓			
Kavallieratos et al. (2020)	Survey	✓	✓	✓	✓	✓	✓	✓	✓		
Kriiaa et al. (2015b)	Survey				✓		✓	✓	✓		
Paul and Rioux (2015)	Survey					✓					
Pietre-Cambacedes and Bouissou (2013)	Survey					✓					
Chockalingam et al. (2017)	Survey	✓								✓	
Abulamddi (2016)	Survey										
Present work	SLR	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

curve to work with a specific tool, initial effort, and costs associated with systemic modeling currently pose a challenge for engineers, hindering the broad implementation of risk assessment activities mentioned in the studies. Also, only a few tools support safety and security co-analysis as native features. Most tools are extended with plugins or used in different approaches, including modeling cyberattacks as external faults. Those approaches represent a need for robust meta-models to support both disciplines and to be used as a base for tools' new features. Adopting such tools with cross-cutting concerns may challenge community acceptance since no standard addresses safety and security combined.

Concerning the application domains, the current research landscape tends towards the transportation and industrial sectors. Standards are already in place for those sectors. However, the reality is that any Internet-connected system is vulnerable to cyberattacks. As emerging technologies increasingly merge the cyber and physical world, potential safety and security concerns must be addressed.

In our future work, we aim to fill the gap in MD methodologies by proposing our own. Despite the significant effort required to realize this proposal, we plan to gradually contribute by establishing a common meta-model for safety and security.

CRedit authorship contribution statement

Victor Luiz Grechi: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration,

Methodology, Investigation, Formal analysis. **André Luiz de Oliveira:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Rosana T. Vaccare Braga:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. List of primary studies

See [Table 9](#).

Appendix B. List of publishers

See [Table 10](#).

Data availability

Data will be made available on request.

Table 9
List of primary studies.

Study ID	Reference	Year	Title
1	Pedroza and Mockly (2020)	2020	Method and Framework for Security Risks Analysis Guided by Safety Criteria
2	Schmittner et al. (2015)	2015	A Case Study of FMVEA and CHASSIS as Safety and Security Co-Analysis Method for Automotive Cyber-Physical Systems
3	Kaneko and Yoshioka (2020)	2020	A Five-Layer Model for the Analysis of Complex Socio-Technical Systems
4	Burton et al. (2012)	2012	Automotive Functional Safety = Safety + Security
5	Castiglione and Lupu (2020)	2020	Hazard Driven Threat Modeling for Cyber Physical Systems
6	Kruck et al. (2021)	2021	Safe and Secure: Mutually Supporting Safety and Security Analyses with Model-Based Suggestions
7	Kriaa et al. (2015a)	2015	A model based approach for SCADA safety and security joint modeling: S-cube
8	Papakonstantinou et al. (2019)	2019	Early Hybrid Safety and Security Risk Assessment Based on Interdisciplinary Dependency Models
9	Liu et al. (2021)	2021	CMSS: Collaborative Modeling of Safety and Security Requirements for Network Protocols
10	Draeger (2015)	2015	A roadmap to a unified treatment of safety and security
11	Zhao et al. (2019)	2019	A Language-Based Multi-View Approach for Combining Functional and Security Models
12	Viet Nguyen et al. (2020)	2020	REAFFIRM: Model-Based Repair of Hybrid Systems for Improving Resiliency
13	Hecht et al. (2020)	2020	Automated Generation of FMEAs using SysML for Reliability, Safety, and Cybersecurity
14	Lyu et al. (2020)	2020	Bayesian Network Based C2P Risk Assessment for Cyber-Physical Systems
15	Fujikura and Kurachi (2020)	2020	A Simultaneous Attack Scenario Generation Method Using the Parallel Behavior Model
16	Tantawy et al. (2022)	2022	Cyber LOPA: An Integrated Approach for the Design of Dependable and Secure Cyber-Physical Systems
17	Ponsard et al. (2021)	2021	A goal-driven approach for the joint deployment of safety and security standards for operators of essential services
18	Saqui-sannes et al. (2021)	2021	Checking sysml models against safety and security properties
19	Martin et al. (2020)	2020	Combined automotive safety and security pattern engineering approach
20	Torkildson et al. (2020)	2020	Improving security and safety co-analysis of STPA
21	Kondeva et al. (2019)	2019	On computer-Aided techniques for supporting safety and security co-engineering
22	Kriaa et al. (2019)	2019	A new safety and security risk analysis framework for industrial control systems
23	Herbert and Kappauf (2021)	2021	A non-parametric cumulative sum approach for online diagnostics of cyber attacks to nuclear power plants
24	Sassone et al. (2019)	2019	A methodology for assuring the safety and security of critical infrastructure based on STPA and Event-B
25	Kumar and Stoelinga (2017)	2017	Quantitative security and safety analysis with attack-fault trees
26	Apvrille and Li (2017)	2017	Safe and Secure Support for Public Safety Networks
27	Beckers et al. (2016)	2016	Standard compliant hazard and threat analysis for the automotive domain
28	Antonino et al. (2018)	2018	A Combined Safety-Hazards and Security-Threat Analysis Method for Automotive Systems
29	Julien (2014)	2014	From Safety Models to Security Models: Preliminary Lessons Learnt
30	Ito (2013)	2013	Finding Threats with Hazards in the Concept Phase of Product Development
31	Macher et al. (2016)	2016	Threat and Risk Assessment Methodologies in the Automotive Domain
32	Utsunomiya et al. (2016)	2016	A Safety Knowledge Representation of the Automatic Driving System
33	Javed et al. (2021)	2021	Safe and secure platooning of Automated Guided Vehicles in Industry 4.0
34	de Souza et al. (2020)	2020	Extending STPA with STRIDE to identify cybersecurity loss scenarios
35	Friedberg et al. (2017)	2017	STPA-SafeSec: Safety and security analysis for cyber-physical systems
36	Carreras Guzman et al. (2021b)	2021	A Comparative Study of STPA-Extension and the UfoI-E Method for Safety and Security Co-analysis
37	Dürrwang et al. (2017)	2017	A Lightweight Threat Analysis Approach Intertwining Safety and Security for the Automotive Domain
38	Reichenbach et al. (2012)	2012	A pragmatic approach on combined safety and security risk analysis
39	Abdo et al. (2018)	2018	A safety/security risk analysis approach of Industrial Control Systems: A cyber bowtie combining new version of attack tree with bowtie analysis
40	Brunel et al. (2014)	2014	A Viewpoint-Based Approach for Formal Safety & Security Assessment of System Architectures
41	Sabaliauskaitė and Mathur (2015)	2015	Aligning Cyber-Physical System Safety and Security
42	Carreras Guzman et al. (2021a)	2021	An integrated safety and security analysis for cyber-physical harm scenarios
43	Cui et al. (2019)	2019	Collaborative Analysis Framework of Safety and Security for Autonomous Vehicles
44	Steiner and Liggesmeyer (2013)	2013	Combination of Safety and Security Analysis - Finding Security Problems That Threaten The Safety of a System
45	Troubitsyna and Vistbakka (2018)	2018	Deriving and formalizing safety and security requirements for control systems
46	Raspotnig et al. (2013)	2013	Enhancing CHASSIS: A Method for Combining Safety and Security
47	Schmittner et al. (2014b)	2014	FMVEA for safety and security analysis of intelligent and cooperative vehicles
48	Howard et al. (2017)	2017	Formal Analysis of Safety and Security Requirements of Critical Systems Supported by an Extended STPA Methodology
49	Jablonski et al. (2022)	2022	Generating Cyber-Physical System Risk Overlays for Attack and Fault Trees using Systems Theory
50	Gallina and Haider (2020)	2020	Making safeconcert security-informed to enable multi-concern modeling
51	Suo et al. (2018)	2018	Merging safety and cybersecurity analysis in product design
52	Kharatyan et al. (2021)	2021	Metamodel for safety and security integrated system architecture modeling
53	Andre et al. (2019)	2019	Parametric analyses of attack-fault trees
54	Hayakawa et al. (2018)	2018	Proposal and application of security/safety evaluation method for medical device system that includes IoT
55	Schwarzl et al. (2021)	2021	Safety and security co-engineering for highly automated vehicles
56	Kriaa et al. (2014)	2014	Safety and security interactions modeling using the bdmp formalism: Case study of a pipeline
57	Martinez et al. (2020)	2020	Safety and security interference analysis in the design stage,
58	Förster et al. (2019)	2019	Safety goals in vehicle security analysesâ€“a method to assess malicious attacks with safety impact
59	Schmittner et al. (2014a)	2014	Security application of Failure Mode and Effect Analysis (FMEA)
60	Kaneko et al. (2020)	2020	STAMP s&s: Safety & security scenario for specification and standard in the society of ai/iot

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Table 9 (continued).

Study ID	Reference	Year	Title
61	Kornecki et al. (2013)	2013	Studying interrelationships of safety and security for software assurance in cyberphysical systems: Approach based on bayesian belief networks
62	Amorim et al. (2017)	2017	Systematic pattern approach for safety and security co-engineering in the automotive domain
63	Temple et al. (2017)	2017	Systems-theoretic likelihood and severity analysis for safety and security co-engineering
64	Hollerer et al. (2021)	2021	Towards a Threat Modeling Approach Addressing Security and Safety in OT Environments
65	Dobaj et al. (2019)	2019	Towards integrated quantitative security and safety risk assessment
66	Witte et al. (2022)	2022	Towards Model Co-evolution Across Self-Adaptation Steps for Combined Safety and Security Analysis
67	Nicklas et al. (2016)	2016	Use case based approach for an integrated consideration of safety and security aspects for smart home applications
68	Rinaldo and Horeis (2020)	2020	A Hybrid Model for Safety and Security Assessment of Autonomous Vehicles
69	Langhendries and Lacaille (2022)	2022	A new model-based risk analysis approach that generate cyberattacks scenarios and combine them with safety risks
70	Liew et al. (2021)	2021	A Novel System-Theoretic Matrix-Based Approach to Analysing Safety and Security of Cyber-Physical Systems
71	Pereira et al. (2019)	2019	A STAMP-based ontology approach to support safety and security analyses.
72	Lamddi (2017)	2017	Developing Dependability Requirements Engineering for Secure and Safe Information Systems with Knowledge Acquisition for Automated Specification
73	Procter et al. (2017)	2017	SAFE and Secure: Deeply Integrating Security in a New Hazard Analysis
74	Popov (2015)	2015	Stochastic modeling of safety and security of the e-motor, an ASIL-D device
75	Roudier and Apvrille (2015)	2015	SysML-Sec: A model driven approach for designing safe and secure systems
76	Li and Hankin (2015)	2015	A Model-based Approach to Interdependency between Safety and Security in ICS
77	Bakirtzis et al. (2022)	2022	An ontological metamodel for cyber-physical system safety, security, and resilience coengineering
78	Sabaliauskaite et al. (2019)	2019	AVES – Automated Vehicle Safety and Security Analysis Framework
79	Sojka et al. (2014)	2014	Case study on combined validation of safety & security requirements
80	Cimatti et al. (2015)	2015	Combining MILS with contract-based design for safety and security requirements
81	Plósz et al. (2017)	2017	Combining Safety and Security Analysis for Industrial Collaborative Automation Systems
82	Quamara et al. (2022)	2022	Formal Analysis Approach for Multi-layered System Safety and Security Co-engineering
83	Hosseini et al. (2023)	2023	Formal Verification of Safety and Security Properties in Industry 4.0 Applications
84	Huang and Kang (2019)	2019	Formal verification of safety & security related timing constraints for a cooperative automotive system
85	Gu et al. (2020)	2020	Integrated Functional Safety and Security Diagnosis Mechanism of CPS Based on Blockchain
86	Sabaliauskaite et al. (2018a)	2018	Integrated safety and cybersecurity risk analysis of cooperative intelligent transport systems
87	Sabaliauskaite et al. (2018b)	2018	Integrating Autonomous Vehicle Safety and Security Analysis Using STPA Method and the Six-Step Model
88	Sabaliauskaite and Adepu (2017)	2017	Integrating six-step model with information flow diagrams for comprehensive analysis of cyber-physical system safety and security
89	Gautham et al. (2020)	2020	Multilevel Runtime Security and Safety Monitoring for Cyber Physical Systems using Model-based Engineering
90	Cui and Sabaliauskaite (2017)	2017	On the Alignment of Safety and Security for Autonomous Vehicles
91	Tekaati et al. (2019)	2019	Potentials for the integration of design thinking along automotive systems engineering focusing security and safety
92	Park and Lee (2019)	2019	Probabilistic safety assessment-based importance analysis of cyber-attacks on nuclear power plants
93	Steiner and Liggesmeyer (2015)	2015	Qualitative and quantitative analysis of CFTs taking security causes into account
94	Katta et al. (2013)	2013	Requirements management in a combined process for safety and security assessments
95	Taguchi et al. (2015)	2015	Safe & sec case patterns
96	Brunel and Chemouil (2015)	2015	Safety and security assessment of behavioral properties using Alloy
97	Monakova et al. (2012)	2012	Security and safety of assets in business processes
98	Bloomfield et al. (2013)	2013	Security-Informed Safety: If It's Not Secure, It's Not Safe
99	de Souza et al. (2019)	2019	STAMP-based approach to analyze safety, security and data privacy
100	Vistbakka and Troubitsyna (2018)	2018	Towards a Formal Approach to Analysing Security of Safety-Critical Systems
101	Cho et al. (2017)	2017	Using Tree-Based Approaches to Analyze Dependability and Security on I&C Systems in Safety-Critical Systems
102	Sahay et al. (2023)	2023	A comparative risk analysis on CyberShip system with STPA-Sec, STRIDE and CORAS
103	Poorhadi et al. (2022)	2022	Analysing the Impact of Security Attacks on Safety Using SysML and Event-B
104	Nurchalifah et al. (2023)	2023	Analysing the Safety and Security of a UV-C Disinfection Robot
105	Bolbot et al. (2023)	2023	A novel risk assessment process: Application to an autonomous inland waterways ship
106	Verma et al. (2018)	2018	A quantitative approach for the likelihood of exploits of system vulnerabilities
107	Dantas and Nigam (2023)	2023	Automating Safety and Security Co-design through Semantically Rich Architecture Patterns
108	Yuan et al. (2023)	2023	Cost-effective maintenance of safety and security barriers in the chemical process industries via genetic algorithm
109	Trentinaglia (2022)	2022	Deriving model-based safety and security assurance cases from design rationale of countermeasure patterns
110	Greiner et al. (2023)	2023	Feature-based software architecture analysis to identify safety and security interactions
111	Sango et al. (2019)	2019	Model-based system, safety and security co-engineering method and toolchain for medical devices design
112	Gallina et al. (2022)	2022	Multiconcern, Dependability-Centered Assurance Via a Qualitative and Quantitative Coanalysis
113	Quamara et al. (2021)	2021	Multi-layered Model-based Design Approach towards System Safety and Security Co-engineering
114	Vistbakka and Troubitsyna (2019)	2019	Pattern-based formal approach to analyze security and safety of control systems
115	Japs and Anacker (2021)	2021	Resolution of safety relevant security threats in the system architecture design phase on the example of automotive industry
116	Martin et al. (2017)	2017	Safety and security co-engineering and argumentation framework

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Table 9 (continued).

Study ID	Reference	Year	Title
117	Japs et al. (2021)	2021	SAVE: Security & safety by model-based systems engineering on the example of automotive industry
118	Macher et al. (2016)	2016	Threat and Risk Assessment Methodologies in the Automotive Domain
119	Meng et al. (2021)	2021	VERDICT: A Language and Framework for Engineering Cyber Resilient and Safe System

Table 10

List of publishers.

Study ID	Publisher subject	Publisher name	Publisher type
18	Aerospace Technology	Journal of Aerospace Information Systems	JOURNAL
26	Communication Technology	Wireless Public Safety Networks 3	BOOK
107	Computer Science in general	ACM Transactions on Cyber-Physical Systems	JOURNAL
97	Computer Science in general	Annual ACM Symposium on Applied Computing	SYMPOSIUM
40	Computer Science in general	CEUR Workshop Proceedings	WORKSHOP
61	Computer Science in general	Federated Conference on Computer Science and Information Systems (FedCSIS)	CONFERENCE
90	Computer Science in general	IARIA CYBER	CONFERENCE
86	Computer Science in general	International Conference on Soft Computing and Intelligent Systems (SCIS)	CONFERENCE
37, 47, 56, 59, 62, 63, 65	Computer Science in general	Lecture Notes in Computer Science (LNCS)	JOURNAL
32, 31	Computer Science in general	Procedia Computer Science	JOURNAL
118	Computer Science in general	Procedia Computer Science	CONFERENCE
39, 102	Cyber Security	Computers & Security	JOURNAL
87	Cyber Security	International Journal on Advances in Security	JOURNAL
76	Cyber Security	International Symposium for ICS & SCADA Cyber Security Research	SYMPOSIUM
44	Dependability	Computer Safety, Reliability, and Security (SAFECOMP) Workshops	WORKSHOP
100	Dependability	European Dependable Computing Conference (EDCC)	CONFERENCE
88	Dependability	IEEE International Symposium on High Assurance Systems Engineering (HASE)	SYMPOSIUM
57, 74	Dependability	International Conference on Computer Safety, Reliability, and Security	CONFERENCE
45	Dependability	International Conference (SAFECOMP)	CONFERENCE
99	Dependability	Latin-American Symposium on Dependable Computing (LADC)	SYMPOSIUM
94	Dependability and security	International Conference on Availability, Reliability and Security (ARES)	CONFERENCE
60	Dependability and security	International Conference on Software Quality, Reliability, and Security (QRS-C)	CONFERENCE
117	Design in general	CIRP Design Conference	CONFERENCE
91	Design in general	Design Society: International Conference on Engineering Design	CONFERENCE
52, 115	Design in general	Proceedings of the Design Society	CONFERENCE
55	Electrical and Information Systems	Elektrotechnik und Informationstechnik	JOURNAL
51	Electrical Power Systems	IET Intelligent Transport Systems	JOURNAL
4, 54, 73	General	ACM International Conference Proceeding Series	CONFERENCE
14, 43, 85	General	IEEE Access	JOURNAL
27	General	Information (Switzerland)	JOURNAL
58	General	Ruhr University Bochum	BOOK
30	General	Springer Berlin, Heidelberg	JOURNAL
29, 28, 23	General	Springer Cham	JOURNAL
83	Industrial automation technology	IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)	CONFERENCE
104	Industrial automation technology	IEEE International Conference on Robotics and Automation (ICRA)	CONFERENCE
79	Industrial automation technology	IEEE International Symposium on Industrial Embedded Systems (SIES)	SYMPOSIUM
64	Industrial automation technology	IEEE International Workshop on Factory Communication Systems - Proceedings (WFCS)	WORKSHOP
111	Medical Technology	Frontiers in Biomedical Devices	CONFERENCE
12	Model-driven software engineering	ACM-IEEE International Conference on Formal Methods and Models for System Design (MEMOCODE)	CONFERENCE
1, 113	Model-driven software engineering	ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)	CONFERENCE
75	Model-driven software engineering	International Conference on Model-Driven Engineering and Software Development	CONFERENCE
109	Model-driven software engineering	International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings	CONFERENCE
103	Model-driven software engineering	International Symposium on Model-Based Safety and Assessment	SYMPOSIUM
114	Model-driven software engineering	Model-Based Safety and Assessment: International Symposium, IMBSA	SYMPOSIUM
77	Model-driven software engineering	Software and Systems Modeling	BOOK
92	Nuclear Technology	Nuclear Engineering and Technology	JOURNAL
9	Parallel and Distributed Processing	IEEE International Symposium on Parallel and Distributed Processing with Applications	SYMPOSIUM
3	Pattern languages	CONFERENCE ON PATTERN LANGUAGES OF PROGRAMS (PLoP)	CONFERENCE
16	Reliability	IEEE Transactions on Reliability	JOURNAL
13, 8	Reliability and Maintainability	Annual Reliability and Maintainability Symposium (RAMS)	SYMPOSIUM
22	Reliability and Risk Analysis	Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability	JOURNAL
105	Reliability and Risk Analysis	Journal of Risk and Reliability	JOURNAL
69	Reliability and Safety	European Conference on Safety and Reliability (ESREL)	CONFERENCE

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Table 10 (continued).

Study ID	Publisher subject	Publisher name	Publisher type
20	Reliability and Safety	European Safety and Reliability Conference	CONFERENCE
19, 36	Reliability and Safety	Reliability Engineering and System Safety	JOURNAL
50	Safety and Risk Analysis	Probabilistic Safety Assessment and Management Conference (PSAM)	CONFERENCE
10	Safety and Security	10th IET System Safety and Cyber-Security Conference	CONFERENCE
81, 80, 89, 96, 95, 93, 106, 116	Safety and Security	Computer Safety, Reliability, and Security (SAFECOMP) Workshops	WORKSHOP
82	Safety and Security	European Dependable Computing Conference	CONFERENCE
7	Safety and Security	IET System Safety and Cyber-Security Conference	CONFERENCE
24	Safety-critical applications	Journal of Critical Computer-Based Systems	JOURNAL
108	Safety in general	Process Safety and Environmental Protection	JOURNAL
42	Safety in general	Safety Science	JOURNAL
49	Security and privacy	ACM Conference on Data and Application Security and Privacy	CONFERENCE
2	Security and privacy	ACM Workshop on Cyber-Physical System Security (CPSS), Part of ASIACCS	CONFERENCE
48	Security and privacy	IEEE European Symposium on Security and Privacy Workshops	SYMPOSIUM
46	Security and privacy	International Conference on Availability, Reliability and Security (ARES)	CONFERENCE
5	Security and privacy	Joint Workshop on CPS and IoT Security and Privacy	WORKSHOP
34, 35, 71	Security and privacy	Journal of Information Security and Applications (JISA)	JOURNAL
66	Self-adaptive systems	Symposium on Software Engineering for Adaptive and Self-Managing Systems	SYMPOSIUM
41	Smart Transportation	Complex Systems Design & Management Asia	CONFERENCE
11	Software Engineering	Asia-Pacific Software Engineering Conference (APSEC)	CONFERENCE
84	Software Engineering	Fundamental Approaches to Software Engineering: International Conference (FASE)	CONFERENCE
25	Software Engineering	IEEE International Symposium on High Assurance Systems Engineering (HASE)	SYMPOSIUM
112	Software Engineering	IEEE Software	JOURNAL
53	Software Engineering	International Conference on Application of Concurrency to System Design	CONFERENCE
72	Software Engineering	Journal of Software Engineering and Applications	JOURNAL
17	Software Engineering	Journal of Software: Evolution and Process	JOURNAL
119	Software Engineering	MDPI Systems	JOURNAL
98	Software Engineering	Software Engineering for Resilient Systems: 5th International Workshop (SERENE)	WORKSHOP
6	Software Reliability Engineering	IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW)	WORKSHOP
21, 38	Software Reliability Engineering	IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW)	SYMPOSIUM
110	Systems Architecture	International Conference on Software Architecture (ICSA)	CONFERENCE
33	Systems Architecture	Journal of Systems Architecture	JOURNAL
101	Systems of Systems Engineering	IEEE Systems Journal	JOURNAL
67	Systems of Systems Engineering	Systems of Systems Engineering Conference (SoSE)	CONFERENCE
70	Telecommunications	Telecom	JOURNAL
68, 78	Vehicular Technology	ACM Computer Science in Cars Symposium	SYMPOSIUM
15	Vehicular Technology	IEEE Vehicular Technology Conference	CONFERENCE

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